



RE:INVENT · L300

Build stateful agentic workflows with Aurora, MCP, and AgentCore

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Alex needs a plan that can survive an interruption

FICTIONAL TRAVEL SCENARIO · LIVE AWS IMPLEMENTATION

A canceled JFK-to-Tokyo trip.
Saved preferences.
A limited package hold.

What must survive
when the worker
disappears?

The screenshot shows a travel planning application interface. At the top, there's a 'Presenter controls' bar with options like 'Projector readability', 'Preview audience layout', and 'Present fullscreen'. Below this is a navigation bar with tabs: 'Concierge', 'Capability ladder', 'Recovery desk', 'System evidence', and 'Solution briefing'. The main content area is titled 'Good evening, Alex.' and 'Where would you like to go?'. It features a section 'A little inspiration for your next chapter' with a photo of a Tokyo street and a listing for 'Tokyo Indie Neighborhood Walk' priced at \$1,599 per traveler. On the right, a 'Your travel brief' section shows a 'Confirmed booking' for 'Tokyo Executive Stopover' with a 'Confirmed' status. Below this, a profile for 'Alex Morgan' is shown. At the bottom, there's a search bar with the text 'Tell me what you have in mind...' and a 'Start with an idea' section with suggestions like 'A quiet wine country escape for two' and 'Help me plan a culture trip to Tokyo'.

Presenter controls
Hidden in fullscreen · Esc to return

Projector readability Preview audience layout Present fullscreen

Concierge Capability ladder Recovery desk System evidence Solution briefing

Meridian live · USD

Good evening, Alex.

Where would you like to go?

Somewhere new. Something familiar. A trip that's yours.

A little inspiration for your next chapter

Tokyo Indie Neighborhood Walk

Yanaka and Shimokitazawa boutique stays, late-morning starts, indie coffee bars, vintage shops, no group tours. Rail pass included.

4 nights · JAL Tours

From **\$1,599** / traveler

Explore this trip →

Confirmed booking

Confirmed

Catalog inventory is booked in Meridian's database. No supplier was contacted and no payment was taken.

Booking record

Alex Morgan
Your preferences, remembered

Flying from **JFK** Next stop **Possibility.**

Travelers **2 adults**

Travel dates **Open to ideas**

Usual budget **\$3,200 per traveler**
\$6,400 for 2 travelers

Start with an idea A quiet wine country escape for two Help me plan a culture trip to Tokyo

Tell me what you have in mind...

Add travelers Change dates Spa & wellness Direct flights

Three questions organize the talk

IMPLEMENTED IN MERIDIAN

Who may act?

Browser access

Workload-to-traveler
grant

Cedar before a tool runs

What must persist?

Traveler facts

Conversation context

Checkpoint + hold intent

How do we prove it?

A new worker

The same thread

One persisted booking

SQL → MCP → retrieval

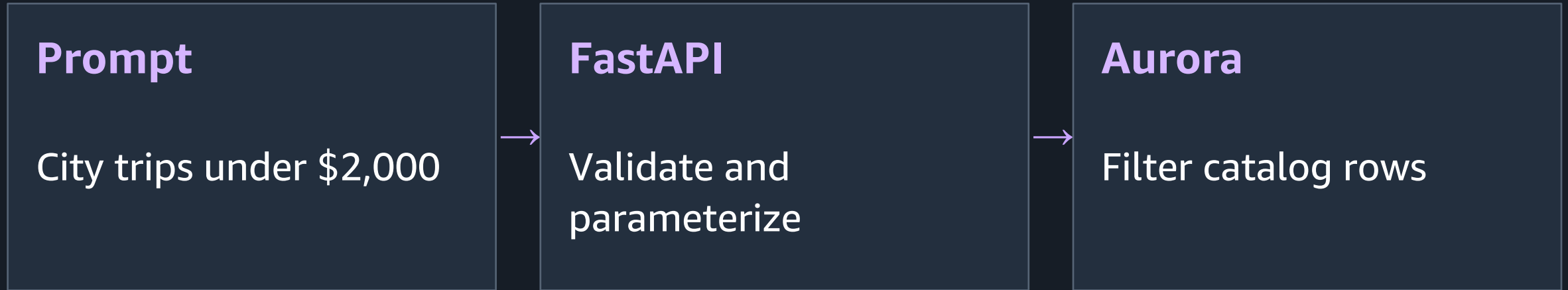
MERIDIAN CAPABILITY LADDER

Add capabilities when the question requires them.



Start with grounded catalog rows

PHASE 1 · DIRECT RDS DATA API

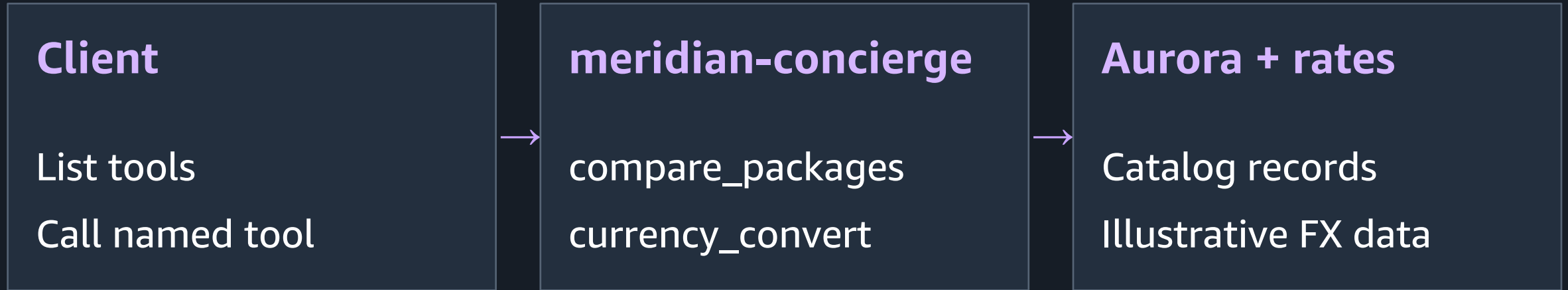


Evidence: executed SQL + returned prices

Then ask for a comparison in euros. The configured phase names its boundary.

Tools make capabilities explicit

PHASE 2 · MCP CONTRACTS



One server for this comparison prompt. PostgreSQL MCP is a separate catalog path.

Search by meaning, then verify

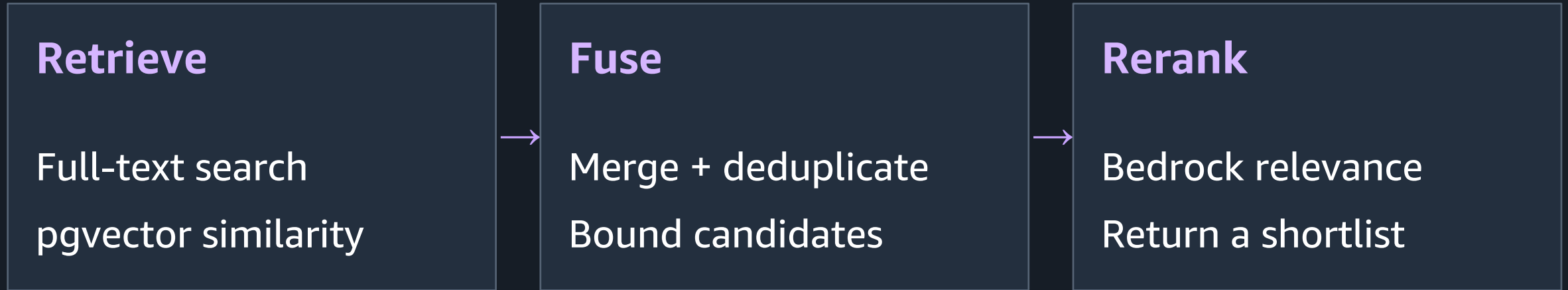
MERIDIAN CAPABILITY LADDER

Retrieval finds candidates. Inventory decides availability.



Words and meaning produce candidates

PHASE 3 · HYBRID RETRIEVAL + RERANK



“Find a quiet, romantic wine-country retreat with a private villa.”

If reranking is unavailable, label the hybrid-order fallback.

Finding a trip does not authorize a reservation

IMPLEMENTED IN MERIDIAN

Search

Interpret the request

Retrieve grounded
candidates

Details

Read package attributes

Check published
durations

Availability

Read inventory

No booking write in
Retrieval

Remember the traveler. Govern the action.

MERIDIAN CAPABILITY LADDER

Context is useful. Authority must be explicit.



Three kinds of state have different owners

IMPLEMENTED IN MERIDIAN

Traveler facts

Aurora preferences

Saved budget

Read under RLS

Conversation

AgentCore Memory

Runtime session context

Aurora turn mirror

Execution

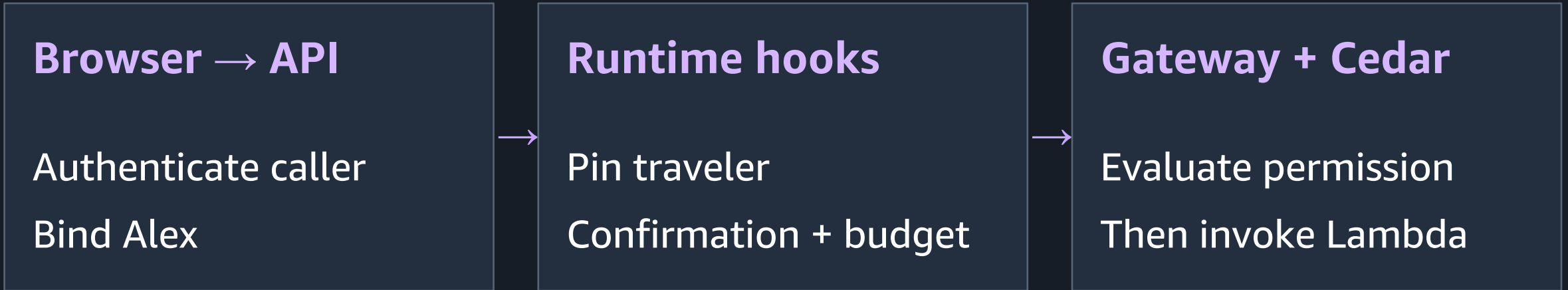
Aurora checkpoints

Worker leases

Stable hold intent

Put authority outside the model

PHASE 4 · TRUST BOUNDARIES



Aurora transaction

Authorize the Lambda workload; apply RLS; validate ownership, price and inventory.

An applicable permit is required

IMPLEMENTED IN MERIDIAN

Read

Search packages

Read package details

Authenticated tools

Hold

Traveler confirmed

≤ 6 travelers · ≤ 12 hours

Within saved budget

Confirm

Traveler confirmed

Within saved budget

Aurora checks owner +
expiry

The worker can disappear

MERIDIAN CAPABILITY LADDER

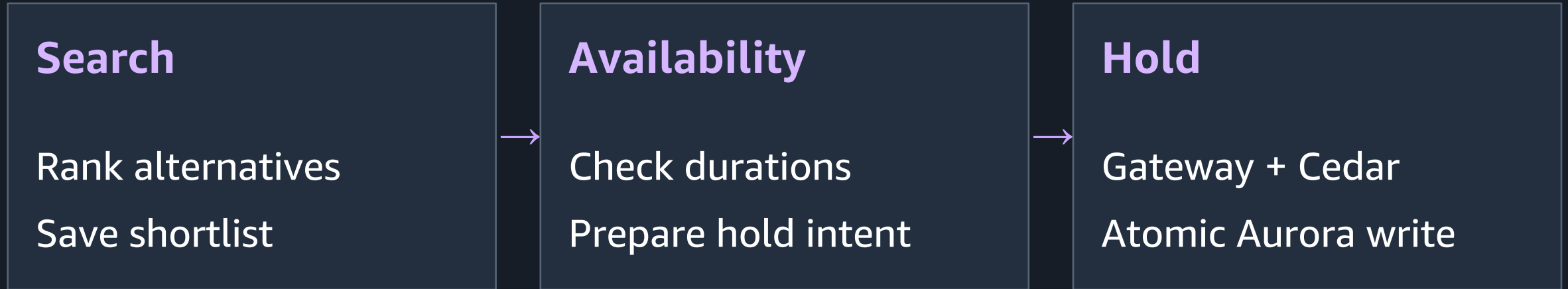
The journey should not.



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Checkpoint progress; transact the business write

PHASE 5 · LANGGRAPH IN FASTAPI



AuroraDataApiSaver stores checkpoints between nodes over HTTPS Data API.

Execution lease → one active worker for a thread → saved intent reused on resume

Pause. Reload. Resume the same journey.

LIVE DEMO · RECORDS, NOT A SIMULATED RESTART

- 1 Start recovery
- 2 Reload the saved URL
- 3 Resume the hold
- 4 Open evidence

A normal pause/resume is not proof that a worker was killed.

The screenshot displays the Amazon Aurora Recovery desk interface. At the top, there are controls for 'Presenter controls' (Hidden in fullscreen · Esc to return), 'Projector readability' (checked), 'Preview audience layout', and 'Present fullscreen'. The main navigation bar includes 'Concierge', 'Capability ladder', 'Recovery desk' (selected), 'System evidence', and 'Solution briefing'. The current journey is 'JFK → Tokyo', with a note 'Original flight canceled - Traveler-reported - inspect itinerary'. The process flow is 'From request to a recorded result', with steps: 1. Find alternatives (5 returned), 2. Save progress (Durable checkpoint), 3. Check the hold (Cedar allowed), and 4. Verify the receipt (Hold recorded). A '15-minute package hold' is shown with '13:50 remaining'. Below this, a section 'Bring it home.' states 'The package is held. Alex confirms the trip with the concierge; the booking policy decides before Aurora confirms.' and includes a 'Take it back to Alex' button. The 'Recommended trip plan' section shows a 'Top catalog match' for 'JAL Premium · Tokyo' with a price of '\$1,949 PER TRAVELER' and details about 'Marunouchi business hotel, Haneda lounge access, car service to...'. The 'Concierge assistance' section shows 'Haneda · hotel assistance' and an 'Airport-area stay shortlist' with options like 'Context recalled' and 'Check lounge options'. On the right, the 'Journey continuity' section lists status updates: 'Traveler authorized', 'Alternatives retrieved', 'Checkpoint persisted', 'Worker interrupted', and 'Saved plan resumed'. At the bottom right, 'Saved journey evidence' is shown with a thread ID and source information.

Three failure windows, three recovery rules

IMPLEMENTED IN MERIDIAN

Before commit

No visible business write
Rollback incomplete
work
Retry from saved state

Commit, reply lost

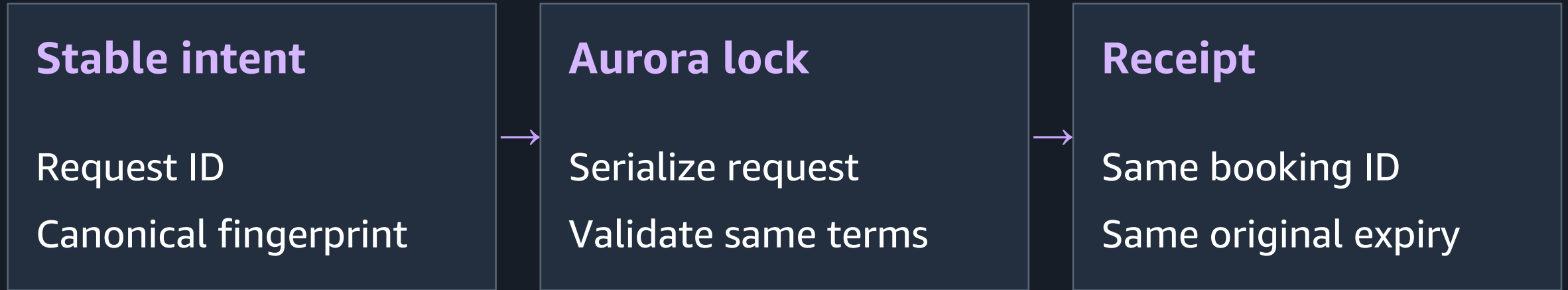
Booking may already
exist
Replay the same request
ID
Return the original
receipt

After checkpoint

Read the saved thread
Wait for the old lease
Continue on a new
worker

The transaction owns replay protection

ONE HOLD PER REQUEST ID · NOT UNIVERSAL EXACTLY ONCE



A reused request ID with different terms is an error.

Inventory locking and lease fencing protect different races.

Follow each claim to an independent record

IMPLEMENTED IN MERIDIAN

Workflow

Thread + checkpoint
Attempt + worker ID
Saved resume receipt

Authority

Workload binding audit
Cedar allow or deny
Pinned arguments

Business result

hold_requests +
bookings
One request, one record
Original expiry retained

The mechanism has deliberate limits

IMPLEMENTED IN MERIDIAN

Single presenter

Shared demo principal
No multi-user login
Trusted Gateway callers

Operational cost

Aurora capacity +
backup
Model calls + memory
Logs and hosted services

Release proof

Local source validated
Runtime source parity
checked
Hosted web still older

Keep these three decisions explicit

IMPLEMENTED IN MERIDIAN

Context

What may the agent recall?

Scope it to the traveler.

Authority

What may the tool do?
Decide before execution.

Durability

What survives the worker?

Persist intent and receipts.

Continue from the source

SOURCE + RUNBOOK + AUTHORITATIVE SERVICE DOCUMENTATION

[github.com/aws-samples/
sample-stateful-agentic-ai-workflows-aurora-mcp-agentcore](https://github.com/aws-samples/sample-stateful-agentic-ai-workflows-aurora-mcp-agentcore)

Presenter path

meridian/DEMO_SCRIPT.md

meridian/docs/PRESENTER_RUNBOOK_2026-09-19.md

AWS documentation: AgentCore Policy · Aurora RDS Data API





Thank you

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 adisamant

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 shayonsanyal

Please complete the session
survey in the mobile app

Presenter preflight and recovery

OPTIONAL · OPERATIONS

Before the room

Correct account + region

Durable health + four tools

Warm one live turn

If a request stalls

Wait up to 55 seconds

Read the saved receipt

Do not invent success

Before leaving

Release only owned tests

Preserve unrelated data

Check retained resources

Temporal policy is a discussion branch

OPTIONAL · DESIGN EXPLORATION

Implemented

Cedar per tool
invocation

Aurora ownership +
expiry

Stable request replay

Explore

History-dependent
policy

Rules across multiple
calls

Additional state
ownership

Do not conflate

A design possibility

An evaluated service

An enabled capability